

Augmented system (per axis)

$$\mathbf{x}[k] = \begin{bmatrix} \delta x_1 & \delta x_2 & \delta x_3 & a_x & a'_x \end{bmatrix}^\top [k], \quad \mathbf{y}[k] = \mathbf{H}[k] \mathbf{x}[k] + \mathbf{r}[k]$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] & dF_{dx}^h[k] \\ 0 & 0 & 0 & 1 & \Delta h_d[k] \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{H}[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -\Delta H_d[k] \end{bmatrix}$$

$$F_{dx}[k] = f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k-i], \quad dF_{dx}^h[k] = (1 - \lambda_c) \sum_{i=1}^d (h_d[k] - h_d[k-i]) f_{dx}[k-i]$$

$$\Delta h_d[k] = h_d[k+1] - h_d[k], \quad \Delta H_d[k] = h_d[k] - h_d[k-d] \quad (\text{desired height, known})$$

a'_x enters only through $\mathbf{F}_e(3,5) = dF_{dx}^h$, $\mathbf{F}_e(4,5) = \Delta h_d$, $\mathbf{H}(2,5) = -\Delta H_d$.

Observability matrix

$$\mathcal{O}_N(k) = \begin{bmatrix} \mathbf{H}[k] \\ \mathbf{H}[k+1] \Phi(k+1, k) \\ \vdots \\ \mathbf{H}[k+N-1] \Phi(k+N-1, k) \end{bmatrix}, \quad \Phi(k+j, k) = \mathbf{F}_e[k+j-1] \cdots \mathbf{F}_e[k], \quad \Phi(k, k) = \mathbf{I}$$

$\text{rank } \mathcal{O}_N(k) = \#\{\text{observable modes}\}; \quad \text{rank} = 5 \Leftrightarrow a'_x \text{ observable over the window.}$

Slope sensitivity (column 5)

$$\mathbf{F}_e[k] \mathbf{e}_5 = \begin{bmatrix} 0 & 0 & dF_{dx}^h[k] & \Delta h_d[k] & 1 \end{bmatrix}^\top$$

$$\mathcal{O}_N(k) \mathbf{e}_5 = \begin{bmatrix} 0 \\ -\Delta H_d[k] \\ 0 \\ \Delta h_d[k] - \Delta H_d[k+1] \\ 0 \\ \Delta h_d[k] + \Delta h_d[k+1] - \Delta H_d[k+2] \\ \vdots \end{bmatrix} \quad \begin{array}{l} y_1 \text{ (odd) rows} = 0 \text{ for } j \leq d; \\ \text{first nonzero } y_1 \text{ at } j=d+1 \propto dF_{dx}^h[k]. \\ \text{every entry } \propto \{\Delta h_d, \Delta H_d\}. \end{array}$$

Structural rank

Sustained hold $\Rightarrow \Delta h_d = \Delta H_d = dF_{dx}^h = 0 \Rightarrow \mathbf{F}_e[k] \mathbf{e}_5 = \mathbf{e}_5$, $\mathbf{H}[k] \mathbf{e}_5 = \mathbf{0} \Rightarrow \mathcal{O}_N \mathbf{e}_5 = \mathbf{0}$:

$$\boxed{\text{rank } \mathcal{O}_N = 5 \iff \Delta h_d \neq 0 \quad (\text{hold: rank} = 4, \ker \mathcal{O}_N = \text{span}\{\mathbf{e}_5\})}$$

a_x stays observable at any fixed height ($\mathbf{H}(2, 4) = 1 \Rightarrow \mathcal{O}_N \mathbf{e}_4 \neq \mathbf{0}$); only a'_x requires motion. The null direction at a hold is exactly $\mathbf{e}_5$: a'_x is the sole unobservable state.

Baseline result; superseded on a hold. The rank-4-on-hold conclusion above is for the *baseline* map $\mathbf{H}(2, 5) = -\Delta H_d$. The companion `5state_aprime_selfmod.tex` restores the actual-height coupling $a_x = a_{\text{det}} - a'_x \delta x$, adding a thermal self-dither to $\mathbf{F}_e(4, 5)$ and $\mathbf{H}(2, 5)$ that is nonzero on a hold $\Rightarrow \text{rank} = 5$ on a hold (weakly, SNR-limited). It also corrects $\mathbf{F}_e(4, 4)$ to 1 here (unknown wall: no a_{det} reversion target).

Observability strength

$\sigma_5(\mathcal{O}_N)$ (weakest mode; its right singular vector $\rightarrow \mathbf{e}_5$ at hold), $\|\mathcal{O}_N \mathbf{e}_5\|^2 = \Delta H_d[k]^2 + (\Delta h_d[k] - \Delta H_d[k+1])^2 +$

$\ \mathcal{O}_N \mathbf{e}_5\ \propto \Delta h_d \text{ (height speed),} \quad \frac{\sigma_5}{\sigma_1} \ll 1 \text{ (} a'_x \text{ weakly observable)}$

Verification

Reconstruct $\mathbf{F}_e[k], \mathbf{H}[k]$ from (f_{dx}, h_d) along a run, form $\mathcal{O}_N(k)$ on a sliding window, report rank, $\sigma_5/\sigma_1, \|\mathcal{O}_N \mathbf{e}_5\|$ versus $|\Delta h_d|$. (`check_observability_5state_aprime.m`)